Quantitative remainders and actual self-dual form supply

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1 A scalar-band remainder for LeBrun's Ricci-energy inequality

23 September 2026. This note concerns the equality geometry of LeBrun's Proposition 4.1: constant-scalar-curvature Kähler metrics, not necessarily Einstein metrics. It derives a global smooth self-dual form, a polynomial Weyl approximation, and differential remainder estimates from that proposition's defect. No Einstein or trace-free-Ricci bound is assumed.

The weighted-monopole identities underlying the argument were already derived in the Round 3 monopole proof, Sections 1–4. Their use below is different from the Einstein-defect argument: the only small energy is the LeBrun defect, and the target includes non-Einstein cscK metrics. The added geometric hypothesis is a two-sided negative scalar-curvature band.

1.1 1. Statement and conventions

Let (M, g) be a connected closed oriented smooth four-manifold. Fix a Spin^c structure with real first Chern class a, and assume that its ordinary unperturbed Seiberg–Witten equations are solvable for every metric conformal to g. The class may be selected separately for each metric; it is held fixed while the conformal weight varies. Set

$$p = (a_g^+)^2 > 0, \qquad -S \leqslant s_g \leqslant -s_0 < 0, \qquad S \geqslant s_0 > 0. \tag{1}$$

This is a metricwise polarized hypothesis, not the assertion that one fixed class is a monopole class for all metrics on M.

Use the exterior-algebra norm on forms and the Hilbert–Schmidt operator norm on $W^+ : \Lambda^+ \to \Lambda^+$. All integrals use g. Define

$$\begin{aligned} &w = \sqrt{6}\,|W^+|, && q = -s + w, && t = s + 2w, && V = \mathrm{Vol}_g(M), \\ &D = \frac{1}{4\pi^2}\int_M \left(\frac{s^2}{24} + \frac{|W^+|^2}{2}\right) - \frac{p}{2}, && \delta = 144\pi^2 D, && \Delta = \int_M q^2 - 72\pi^2 p, \\ &K = \frac{S}{\sqrt{s_0}}, && C_0 = \left(1 + \frac{3S}{2s_0}\right)^2. \end{aligned} \tag{2}$$

Here $q \geqslant s_0$. We will prove, not assume, that

$$\boxed{\delta = \Delta + \frac{1}{2}\int_M t^2, \qquad \Delta \geqslant 0.} \tag{3}$$

Theorem. For every $\eta > 0$ there is a smooth positive weight f, a smooth weighted Seiberg–Witten solution (A, Φ), and its smooth self-dual two-form

$$D_A\Phi = 0, \qquad -iF_A^+ = f\sigma(\Phi), \qquad \psi = 2\sqrt{2}\,\sigma(\Phi), \qquad \rho = |\Phi|^2 = |\psi| \leqslant K, \tag{4}$$

such that all the following estimates hold simultaneously. Although ψ may vanish, $\rho = |\Phi|^2$ is smooth, so no nonsmooth norm derivative is involved:

$$\begin{aligned} \int |d\rho|^2 &\leqslant \frac{2\Delta}{27} + \eta, \quad \int |d\psi|^2 \leqslant \frac{2\Delta}{27} + \eta, \\ \int q\left(\rho - \frac{2}{3}\sqrt{q}\right)^2 &\leqslant \frac{4\Delta}{9} + \eta, \qquad \int E_\lambda \leqslant \frac{2\Delta}{27} + \eta, \end{aligned} \tag{5}$$

where

$$E_\lambda = \langle W^+\psi, \psi\rangle + \frac{w}{3}\rho^2 \geqslant 0. \tag{6}$$

In (5), $d\psi$ is the exterior derivative, not the covariant derivative. Define a smooth trace-free symmetric endomorphism of Λ^+ by

$$\mathscr{B}_\psi = \frac{\rho^2}{8} I - \frac{3}{8}\psi \otimes \psi. \tag{7}$$

Then

$$\boxed{\int |W^+ - \mathscr{B}_\psi|^2 \leqslant \frac{1}{27}\int t^2 + \left(\frac{C_0}{27} + \frac{1}{18}\right)\Delta + \eta \leqslant \frac{2C_0 + 7}{54}\,\delta + \eta.} \tag{8}$$

The full covariant derivative satisfies the stronger retained identity estimate

$$\begin{aligned} \int |\nabla\psi|^2 &\leqslant \frac{4\Delta}{27} - \frac{1}{3}\int t\rho^2 + \eta \\ &\leqslant \frac{4\Delta}{27} + \frac{K^2}{3}\sqrt{V\int t^2} + \eta \\ &\leqslant \frac{4\delta}{27} + \frac{K^2}{3}\sqrt{2V\delta} + \eta. \end{aligned} \tag{9}$$

Let H denote the harmonic self-dual representative of a_g^+, with the curvature convention $\mathrm{proj}_{\mathscr{H}^+}(iF_A^+) = 2\pi H$. There is also actual harmonic-projection control:

$$\left\|\sqrt{q}\,\psi + 4\sqrt{2}\pi H\right\|_2^2 \leqslant \frac{4\Delta}{9} + \eta. \tag{10}$$

If $\lambda_1(g) > 0$ is the first positive scalar Laplace eigenvalue, then

$$\begin{aligned}\int (s-\bar{s})^2 &\leqslant \frac{1}{3}\int t^2 + \left(\frac{4C_0}{3} + \frac{2K^2}{\lambda_1(g)}\right)\Delta \\ &\leqslant \left(\frac{2}{3} + \frac{4C_0}{3} + \frac{2K^2}{\lambda_1(g)}\right)\delta.\end{aligned} \tag{11}$$

The last estimate is about g alone and has no η: apply the preceding construction for arbitrarily small η. Uniform scalar oscillation control along degenerating metrics requires control of the displayed spectral factor; none is implicit here.

If $D = 0$, the weight can be taken exactly as $f = \sqrt{q}$, and all slack terms vanish. The metric is negative cscK, with the complex structure induced by its nonvanishing positive spinor. No conclusion $\mathrm{Ric}^\circ = 0$ follows.

1.2 2. Weighted solvability and a derivative-free amplitude bound

For any smooth positive f, solve the ordinary equations for $\widehat{g} = f^{-2}g$. Write $\beta : S^+_g \to S^+_{\widehat{g}}$ for the fiberwise isometric conformal spinor identification. The spinor on g is

$$\Phi = f^{-3/2}\beta^{-1}\widehat{\Phi}. \tag{12}$$

Dirac conformal covariance gives $D^g_A\Phi = 0$. As differential two-forms the unscaled quadratic map obeys $\sigma_{\widehat{g}}(\beta\zeta) = f^{-2}\sigma_g(\zeta)$. Consequently $\sigma_g(\Phi) = f^{-1}\sigma_{\widehat{g}}(\widehat{\Phi})$. The self-dual splitting of two-forms is conformally invariant, giving (4). This calculation keeps separate the bundle identification, the spinor weight, and the two-form weight.

Normalize $|\sigma(\Phi)|^2 = |\Phi|^4/8$. The spinor Weitzenböck equation is

$$4\nabla^*_A\nabla_A\Phi + (s + f\rho)\Phi = 0. \tag{13}$$

At a maximum of ρ, this implies $f\rho \leqslant -s$. Choose smooth $f_j \geqslant \sqrt{s_0}$ converging uniformly to $\sqrt{q}$. Such approximants exist on the fixed compact manifold, including when $|W^+|$ is nonsmooth at its zeros. Then every solution chosen for f_j has $\rho_j \leqslant S/\sqrt{s_0} = K$, independently of all derivatives of f_j.

1.3 3. Four nonnegative defects and the total remainder

For a smooth weight and its solution define

$$\begin{aligned} D_{\mathrm{amp}} &= 2\int |d\rho|^2, \\ D_{\mathrm{quad}} &= \int (4\rho|\nabla_A\Phi|^2 - |\nabla\psi|^2), \\ D_H &= \int (|d\psi|^2 + |d^*\psi|^2) = 2\int |d\psi|^2, \\ D_{\mathrm{eig}} &= 2\int E_\lambda, \qquad D_0 = D_{\mathrm{amp}} + D_{\mathrm{quad}} + D_H + D_{\mathrm{eig}}. \end{aligned} \tag{14}$$

Each term is nonnegative. For D_{quad}, the quadratic map is the parallel Hopf map, whose differential satisfies

$$|d\psi_z(v)|^2 = 4|z|^2|v|^2 - 4(\mathrm{Im}\langle z, v\rangle)^2 \leqslant 4|z|^2|v|^2. \tag{15}$$

For D_{eig}, the least eigenvalue of a trace-free symmetric three-by-three matrix is at least $-\sqrt{2/3}|W^+| = -w/3$.

Pair (13) with $\rho\Phi$ and integrate by parts:

$$2\int |d\rho|^2 + 4\int \rho|\nabla_A\Phi|^2 + \int s\rho^2 + \int f\rho^3 = 0. \tag{16}$$

The self-dual two-form Weitzenböck formula reads

$$D_H = \int \left(|\nabla\psi|^2 + \frac{s}{3}\rho^2 - 2\langle W^+\psi, \psi\rangle\right). \tag{17}$$

Combining (16) and (17) gives the exact identity

$$D_0 = \int \left(\frac{2}{3}q\rho^2 - f\rho^3\right). \tag{18}$$

Let $u = 3\rho/2$ and

$$\begin{aligned} &P = \int (qu^2 - fu^3) = \frac{27}{8}D_0, \quad X = \left(\int q^3 f^{-2}\right)^{1/3}, \\ &Y = \left(\int f^4\right)^{1/3}, \qquad\qquad Z = \int fu^3, \qquad C = \int f^2u^2. \end{aligned} \tag{19}$$

Hölder gives two nonnegative remainders

$$H_1 = XZ^{2/3} - \int qu^2, \qquad H_2 = YZ^{2/3} - C. \tag{20}$$

Hodge orthogonality and the actual curvature equation give

$$C = 18\|F_A^+\|_2^2 = 72\pi^2 p + 18J, \qquad J = \|iF_A^+ - 2\pi H\|_2^2. \tag{21}$$

Here $p > 0$ implies $Z > 0$. Direct factorization yields

$$YX^2 - 72\pi^2 p = 18J + H_2 + Y\frac{X + Z^{1/3}}{Z^{2/3}}(P + H_1). \tag{22}$$

First suppose $f = \sqrt{q}$ is smooth. Put $E = \int q^2$, $\theta = (Z/E)^{1/3}$. Then $X = Y = E^{1/3}$, $H_1 = H_2 = H$, and $0 < \theta \leqslant 1$. Thus

$$\Delta = 18J + H + \frac{1+\theta}{\theta^2}(P+H), \qquad J \leqslant \frac{\Delta}{18}, \quad P \leqslant \frac{\Delta}{2}, \quad D_0 \leqslant \frac{4\Delta}{27}. \tag{23}$$

This proves the differential and eigenvalue assertions in (5).

For amplitude control use the probability measure $d\nu = q^2 d\mu/E$, set $r = u/\sqrt{q}$, and write $c = \int r^2 d\nu$, $z = \int r^3 d\nu$, $m = \int r\, d\nu$. The inequalities above imply $z \leqslant c \leqslant z^{2/3}$, so $c \leqslant 1$. Cauchy–Schwarz gives $c^2 \leqslant mz \leqslant mc$, hence $m \geqslant c$. Therefore

$$\int q(u - \sqrt{q})^2 = E(c - 2m + 1) \leqslant E - C \leqslant \Delta. \tag{24}$$

Finally $f\psi = -2\sqrt{2}\, iF_A^+$; (21) gives (10).

For the general approximants f_j, (18)–(22) remain exact. Their left side in (22) tends to Δ. If $\vartheta_j = Z_j^{1/3}/X_j$, then $0 < \vartheta_j \leqslant 1$ and the last coefficient is at least $2Y_j/X_j \to 2$. Hence (23) holds with errors tending to zero. For (24), retain the fixed measure $q^2 d\mu/E$. Uniform convergence of the weights and $u_j \leqslant 3K/2$ give $z_j \leqslant c_j + o(1)$, while Hölder still gives $c_j \leqslant z_j^{2/3}$. Moreover $c_j \geqslant 72\pi^2 p/E - o(1) > 0$. It follows that $c_j \leqslant 1 + o(1)$ and $m_j \geqslant c_j - o(1)$, proving (24) with vanishing error. Harmonic-projection control follows in the same way from the exact curvature equation. Choose a single sufficiently large j to make every error below any prescribed allowance.

No convergence of spinors, uniform derivative bounds for the weights, or common smoothing rate along varying geometries is used. All asserted forms are smooth actual weighted-monopole forms, not formal algebraic choices of a Weyl eigenline.

1.4 4. The defect split and polynomial Weyl approximation

The pointwise scalar identity is

$$\frac{s^2}{24} + \frac{|W^+|^2}{2} = \frac{q^2}{36} + \frac{t^2}{72}. \tag{25}$$

Integration gives (3); $\Delta \geqslant 0$ was established in (22)–(23).

For any ψ, without assuming it is an eigenvector or is nonzero, trace-freeness of W^+ gives

$$\boxed{|W^+ - \mathscr{B}_\psi|^2 = \frac{1}{6}\left(w - \frac{3}{4}\rho^2\right)^2 + \frac{3}{4}E_\lambda.} \tag{26}$$

For example, $|\mathscr{B}_\psi|^2 = 3\rho^4/32$ and $-2\langle W^+, \mathscr{B}_\psi\rangle = 3\langle W^+\psi, \psi\rangle/4$; completing the square proves (26). This avoids eigenline choices and division by spectral gaps.

Set $a_0 = q - u^2$. Because $q \geqslant s_0$ and $u \leqslant 3K/2$,

$$\int a_0^2 \leqslant C_0 \int q(u - \sqrt{q})^2 \leqslant C_0\Delta + \eta. \tag{27}$$

The relation $w - 3\rho^2/4 = (t + a_0)/3$, (26), (27), and (5) give (8). All approximation errors can be decreased in advance to absorb their fixed coefficients into the displayed η.

1.5 5. Covariant derivative, scalar oscillation, and equality

Rearranging (17) using (6) yields the exact identity

$$\int |\nabla\psi|^2 = D_H + D_{\mathrm{eig}} - \frac{1}{3}\int t\rho^2. \tag{28}$$

The two nonnegative terms on the right are bounded jointly by D_0, not merely separately. Equations (23), (3), and Cauchy–Schwarz prove (9). In particular the unconditional conclusion has order $\delta + \sqrt{V\delta}$ with the displayed scalar-band factor; it is not a linear $C\delta$ estimate. If the additional sign $t \geqslant 0$ holds, (28) does give $\int |\nabla\psi|^2 \leqslant 4\Delta/27 + \eta$.

To prove (11), Poincaré and $0 \leqslant \rho \leqslant K$ give

$$\int (\rho^2 - \bar{\rho}^{\,2})^2 \leqslant 4K^2 \int (\rho - \bar{\rho})^2 \leqslant \frac{8K^2}{27\lambda_1(g)}\Delta + \eta. \tag{29}$$

Use $s = t/3 - 3\rho^2/2 - 2a_0/3$, subtract the constant $-3\bar{\rho}^{\,2}/2$, and apply $(x+y+z)^2 \leqslant 3(x^2+y^2+z^2)$. The mean $\bar{s}$ minimizes the squared scalar error over constants. Equations (27) and (29) give (11) after letting the auxiliary allowance tend to zero.

When $\delta = 0$, (3) gives $t = 0$, so $w = -s/2 > 0$ and $q = -3s/2$ are smooth. Use the exact weight $\sqrt{q}$. Now $D_0 = 0$, $d\rho = 0$, $\rho = 2\sqrt{q}/3 > 0$, and (28) gives $\nabla\psi = 0$. Thus $\sqrt{2}\psi/\rho$ is a parallel self-dual form of norm $\sqrt{2}$, defining a Kähler structure with the chosen orientation. Its scalar curvature is constant and negative. The everywhere nonzero positive spinor gives the compatible Spin^c structure. This is the classical equality geometry, now visible directly in the retained defects.

An unequal-curvature product of closed hyperbolic surfaces is an essential check. With factor curvatures $-\alpha, -\beta$, $\alpha, \beta > 0$, its compatible Kähler class has $p = (\alpha+\beta)^2 V/(8\pi^2)$, whereas $s = -2(\alpha+\beta)$, $w = \alpha+\beta$, and $D = 0$. But $|\mathrm{Ric}^{\circ}|^2 = (\alpha-\beta)^2$, which need not vanish. Therefore this theorem supplies no unqualified Einstein or trace-free-Ricci remainder.

1.6 6. Supplying a second-order budget

The preceding theorem has no second-order hypothesis. Such a budget can also be obtained if one supplies integral Ricci and determinant-curvature control. This step uses the bundle Hessian estimate established in `round4_rigidity.md`, Section 2, and rederived in `round21_analytic.md`, Section 2:

$$\begin{aligned} \|\nabla^2 u\|_2^2 &\leqslant 2\|\nabla^*\nabla u\|_2^2 + 32\|u\|_\infty^2 \|\mathrm{Ric}\|_2^2 + 33\|u\|_\infty^2 \|\mathscr{F}\|_2^2, \\ \|\nabla u\|_4^2 &\leqslant 4\|u\|_\infty \|\nabla^2 u\|_2. \end{aligned} \tag{30}$$

Here $\mathscr{F}$ is the compatible bundle connection curvature, with component norm $\sum_{ij} |\mathscr{F}_{ij}|_{\mathrm{HS}}^2$. Equation (30) is an integration-by-parts estimate on a closed manifold; it assumes no Sobolev constant or differentiated-curvature bound.

Second-order corollary. In addition to (1), assume

$$⨍ |\mathrm{Ric}|^2 \leqslant R_0, \qquad \frac{(-a^2)_+}{V} \leqslant A_0, \qquad \frac{D}{V} \leqslant D_0^*. \tag{31}$$

For every allowance $\eta > 0$, the solution in the theorem may be chosen so that it also satisfies

$$\boxed{\fint \left(|\nabla^2 \psi|^2 + |\Delta_H \psi|^2 \right) \leqslant C(S, s_0, R_0, A_0, D_0^*).} \tag{32}$$

The constant is independent of volume, injectivity radius, Sobolev constants, and derivatives of the approximating weight.

Proof. To make the dependence explicit, put

$$H_W = \frac{S^2 + 288\pi^2 D_0^*}{12}, \qquad Q_1 = S + \sqrt{6 H_W}. \tag{33}$$

Since $w = (t-s)/2$, (3) gives $\fint |W^+|^2 \leqslant H_W$ and $\fint q \leqslant Q_1$. Choose the smooth approximants sufficiently close that $f_j^2 \leqslant q + s_0$, in addition to $f_j \geqslant \sqrt{s_0}$. The weighted curvature equation and $\rho \leqslant K$ give

$$\fint |F_A^+|^2 \leqslant \frac{K^2}{8}(Q_1 + s_0), \qquad \fint |F_A|^2 = 2 \fint |F_A^+|^2 - \frac{4\pi^2 a^2}{V} \leqslant \frac{K^2}{4}(Q_1 + s_0) + 4\pi^2 A_0. \tag{34}$$

The curvature of the positive-spinor connection satisfies the dimensional representation estimate

$$|\mathscr{F}^{S^+}|^2 \leqslant C\big(|W^+|^2 + s^2 + |\operatorname{Ric}^\circ|^2 + |F_A|^2 \big). \tag{35}$$

The W^- block acts trivially on the positive-spin representation; it does not enter this estimate. Hence its normalized integral is bounded by a constant depending only on the parameters in (31). Equation (13) and $|\Phi|^2 \leqslant K$ give

$$\fint |\nabla_A^* \nabla_A \Phi|^2 \leqslant \frac{K}{8} \left[S^2 + K^2 (Q_1 + s_0) \right]. \tag{36}$$

Apply (30) to Φ, whose supremum norm is at most $\sqrt{K}$. Equations (31), (35), and (36) bound $\fint |\nabla_A^2 \Phi|^2$. The second inequality in (30) then gives

$$\fint |\nabla_A \Phi|^4 \leqslant 16K \fint |\nabla_A^2 \Phi|^2. \tag{37}$$

Twice differentiating the parallel quadratic map, rather than differentiating its PDE or the weight, gives

$$|\nabla^2 \psi|^2 \leqslant C\big(K |\nabla_A^2 \Phi|^2 + |\nabla_A \Phi|^4 \big). \tag{38}$$

Finally use $\Delta_H \psi = \nabla^* \nabla \psi + (s/3)\psi - 2W^+ \psi$, $|\nabla^* \nabla \psi|^2 \leqslant 4 |\nabla^2 \psi|^2$, and $|\psi| \leqslant K$. This proves (32). The bounds hold uniformly for all sufficiently close approximants, so selecting one that meets the first-order allowances causes no further loss. In particular the proof never assumes a bound on df_j.

1.7 7. The realizable negative-Einstein Weyl branch

The same actual form supplies the *allowed* branch in the Round 21 two-form criterion. This is a form-production result, not an additional Einstein obstruction.

Set

$$\omega = \frac{\sqrt{3}}{2}\psi, \qquad A_{\omega,1/2} = \frac{1}{2}\left(\frac{|\omega|^2}{3} I - \omega \otimes \omega\right) = \mathscr{B}_\psi, \qquad r = \fint |\operatorname{Ric} + g|^2. \tag{39}$$

Writing $e = s + 4$ and retaining $a_0 = q - 9\rho^2/4$, one has the exact amplitude identity

$$|\omega|^2 - 2 = \frac{t}{6} - \frac{e}{2} - \frac{a_0}{3}. \tag{40}$$

Since $|e|^2 \leqslant 4|\operatorname{Ric} + g|^2$, it follows that

$$\int (|\omega|^2 - 2)^2 \leqslant \frac{1}{12}\int t^2 + 3\int |\operatorname{Ric} + g|^2 + \frac{C_0}{3}\Delta + \eta. \tag{41}$$

Consequently the zeroth-order calibration defect obeys

$$\begin{aligned} d_{1/2} &:= \fint \left(|\operatorname{Ric} + g|^2 + |W^+ - A_{\omega,1/2}|^2 + (|\omega|^2 - 2)^2\right) \\ &\leqslant 4r + \frac{13}{108V}\int t^2 + \left(\frac{10C_0}{27} + \frac{1}{18}\right)\frac{\Delta}{V} + \eta \\ &\leqslant 4r + \frac{10C_0 + 8}{27}\frac{\delta}{V} + \eta. \end{aligned} \tag{42}$$

Here the free approximation allowance was chosen to be ηV before division by volume. Under (31), (32) also provides $\fint |\Delta_H \omega|^2 \leqslant C(S, s_0, R_0, A_0, D_0^*)$. Thus both the bounded calibration and its second-order source budget are supplied by actual monopole equations.

A common sufficient condition for the negative-square budget is

$$a^2 \geqslant T := 2\chi(M) + 3\tau(M). \tag{43}$$

It must be verified for the selected class; it does not follow merely from conformal-class solvability. With the current norm conventions, Chern–Weil gives

$$4\pi^2 T = \int \left(2|W^+|^2 + \frac{s^2}{24} - \frac{|\operatorname{Ric}^\circ|^2}{2}\right), \qquad \frac{(-a^2)_+}{V} \leqslant \frac{r}{8\pi^2}. \tag{44}$$

Also $\fint |\operatorname{Ric}|^2 \leqslant 8 + 2r$. Therefore, in a fixed scalar band, bounds on r and D/V, together with (43), are sufficient for all the budgets in this corollary. There is no separate uncontrolled negative-square parameter in that special application.

At $r = D = 0$, the form has norm $\sqrt{2}$, is parallel, and yields the Kähler–Einstein Weyl amplitude $t_{\text{branch}} = 1/2$ in the notation of Round 21. This branch is realizable. For small positive defects, the

argument yields the displayed form and curvature estimates, not a global metric correction to an exact Einstein metric.

1.8 8. Scaling, relation to prior results, and use

Under $g \mapsto cg$, the quantities s, w, q, S, s_0 scale by c^{-1}, K by $c^{-1/2}$, V by c^2, and λ_1 by c^{-1}. The numbers $\delta, \Delta, C_0, K^2\sqrt{V}, K^2/\lambda_1$ are unchanged. Transporting the weighted solution rescales its differential two-form ψ by $c^{1/2}$, consistently with every displayed energy.

The theorem requires neither W^- control nor a uniform Sobolev, injectivity-radius, volume, or diameter bound. It does require the scalar band and actual weighted-monopole solvability. These hypotheses are not consequences of merely specifying a cohomology vector.

Relative to Round 17, the new control parameter is D_{LB} alone: it retains the full cscK equality set and uses no Ricci-error input. Relative to `round21_analytic.md`, the form here is produced by actual monopole equations. Sections 6–7 state the additional hypotheses that supply its Hodge-Laplacian budget; the first-order theorem alone must not be used as if that budget were automatic. Pointwise near-negative-Einstein metrics provide a scalar band, but small LeBrun defect remains a separate condition to be established.

The useful new remainder is (8), together with the actual global form and its differential controls (5), (9), and (10). It does not assert closeness of metrics to a global Kähler metric, compactness of a degenerating family, or exclusion of all Einstein metrics on the smooth manifold. Those statements require additional arguments.

1.8.1 Sources

- Claude LeBrun, *Ricci Curvature, Minimal Volumes, and Seiberg–Witten Theory*, Proposition 4.1 and Proposition 4.2, original paper.
- Claude LeBrun, *Four-manifolds, curvature bounds, and convex geometry*, Lemma 3.7, weighted equations.
- `round3_monopole_input.md`, Sections 1–4: quadratic-map normalization, four retained defects, Hölder remainders, and smooth approximation.
- `round17_integral.md`: earlier arbitrary-scalar structural estimates with different curvature/Ricci and topological error inputs.
- The scalar-band hypotheses, cscK polynomial fit, and estimates (8)–(11) are established in this note. The finite symbolic check `line1_scalar_band_check.py` checks algebra and the unequal-product calibration; it does not verify weighted solvability or the analytic existence hypothesis.